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Hackathon Use Case 1

AI Powered Intelligent Ticket Routing & Resolution Agent

Problem Statement

IT services companies receive thousands of support tickets daily:

- Misrouted tickets
- Slow resolution
- Poor categorization/ classification
- Manual triaging

Build:

An AI system that:

1. Classifies incoming tickets
2. Routes to correct department
3. Suggests resolution wherever possible
4. Escalates when uncertain
5. Tracks confidence score

Technical Requirements

Classification Layer

- Use embeddings or fine-tuned model
- Categorize into:
    - Infrastructure
    - Application
    - Security
    - Database
    - storage
    - Network

- Access Management

RAG Layer

- Retrieve similar past tickets
- Suggest resolution steps

Agentic Layer (Bonus)

- If confidence < threshold → escalate
- If repeated issue → suggest automation

Evaluation

- Accuracy & F1 Score
- Solution design
- Functional Value and Usability (ease of use)
- Feasibility
- Security
- innovation
- Semantic similarity scoring
- LLM-as-judge evaluation

Judging Criteria

1. Technical Soundness of approach

Evaluate the depth and quality of the underlying AI and engineering implementation.

- Accuracy and F1 score of proposed solution
- Use of Open-Source Models & Tools
- Code Quality & Modularity: Clean architecture, reusable components and well-documented code.
- Performance & Efficiency: How optimized is the solution in terms of processing speed and memory usage?

2. Functional Value & Usability

Assess how well the solution works in real-world scenarios.

- Robustness to Document Variants: Can the solution handle logs, traces and metrices and how is causal analysis attempted

- Preservation of Utility: Does it redact only sensitive information while retaining the rest of the document's structure and content?
- Causal analysis and correlation with symptoms.
- Ease of Use / Integration: Can the solution be deployed or used via a simple interface (CLI, Web UI) or API?

3. Problem Understanding & Approach

Evaluate how deeply the team understood and innovatively tackled the challenge.

- Clarity in Problem Framing:
- Innovativeness in Approach: Any creative use of models, hybrid techniques, or pre/post-processing strategies?
- Scalability & Generalizability: Can the solution be adapted to other document types or languages easily?
- Open-Source Readiness: Is the project properly licensed, documented, and structured for future contributions?

Bonus Points

- Real-world Relevance: Demonstrated how the solution could be applied in other domains (transfer learning) for cause to symptom analysis
- Team Collaboration & Presentation: Clarity of explanation, demo quality, and division of roles.
- Ethical Awareness: Consideration of biases, privacy risks, or limitations in their design.

Impact Assessment

1. Data Privacy and Compliance

- Impact: Promotes the adoption of AI tools that align with global and national data protection laws such as India's DPDP Act, GDPR, and HIPAA.
- Assessment Metric: Number of solutions that demonstrate accurate, auditable redaction of personally identifiable information (PII) across multiple document types.
- Outcome: Enables safer data sharing in sectors like healthcare, education, governance, and finance without risking individual privacy.

2. Open-Source Innovation and Community Building

- Impact: Fosters a growing ecosystem of open-source privacy tech by encouraging participants to publish reusable, modular, and well-documented code.
- Assessment Metric: Number of GitHub repositories with open licenses, contributions post-hackathon, forks/stars, and interest from other developers.
- Outcome: Reduces entry barriers for startups, public sector agencies, and researchers to adopt deidentification pipelines without vendor lock-in.

3. Skill Development in Responsible and Applied AI

- Impact: Builds deep capabilities in applying AI/ML for real-world, socially impactful use cases especially in computer vision, layout analysis, and responsible AI practices.
- Assessment Metric: Diversity and quality of models/techniques used (e.g., LayoutLM, spaCy NER), number of first-time contributors to privacy tech, and skill feedback from participants.
- Outcome: Cultivates a new generation of AI practitioners fluent in balancing performance and privacy, critical for future workforce readiness.

4 Cross-Domain Applications

- Impact: Creates plug-and-play tools for redacting data in government forms, health records, legal documents, and other sensitive formats.
- Assessment Metric: Number of use-case demos presented, feasibility of integration into real-world workflows (e.g., APIs, dashboards), and interest from domain stakeholders.
- Outcome: Accelerates the digitization of services with embedded privacy safeguards, enhancing trust in AI-driven automation.

Data Sources (Free & Practical)

Option A – Kaggle Datasets

- IT Support Ticket Dataset
- Helpdesk Ticket Classification Dataset
- ServiceNow-like datasets

Option B – StackOverflow Dataset

- Filter by tags

- Convert Q&A into ticket-resolution format

Option C – Synthetic Enterprise Dataset (Recommended)

Provide:

- 1,000 synthetic tickets (CSV)
  - title
  - description
  - category
  - resolution
  - priority

Generate using LLM to simulate realistic enterprise patterns.